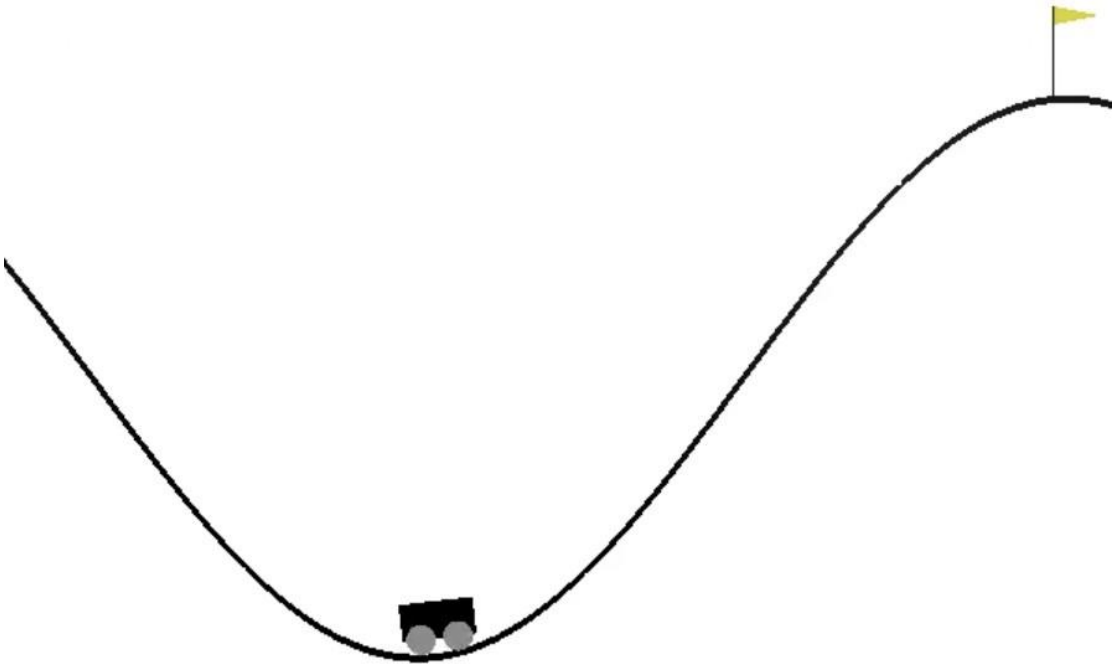
Report of HW1: Implement DQN in MountainCar-v0



gym environment config

Observation

Num	Observation	MIN	MAX
0	Car Position	-1.2	0.6
1	Car Velocity	-0.07	0.07

```
Box([-1.2  -0.07], [0.6  0.07], (2,), float32)
```

Action

Num	0	1	2
Action	Accelerate to the Left	Don't accelerate	Accelerate to the Right

Reward

$$f(x) = \begin{cases} 0, & \text{agent position} \geq 0.5 \text{ on top of the mountain} \\ -1, & \text{elsewhere} \end{cases}$$

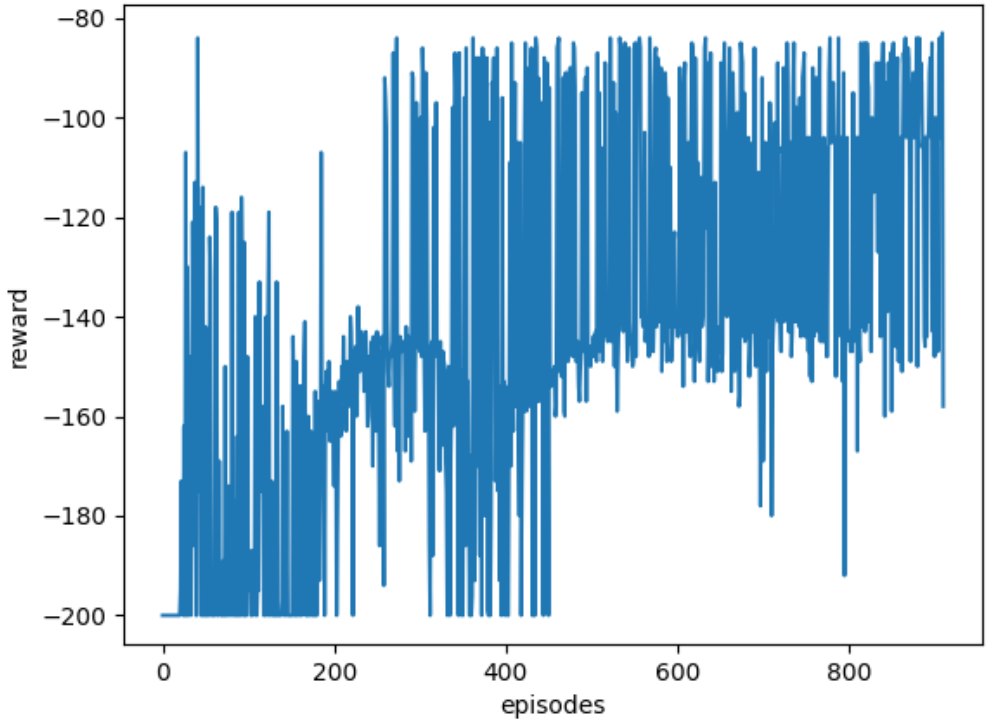
Parameters

Runtime environment	Windows 10 + Python 3.7.9 + torch 1.8.2
CPU	Intel(R) Core(TM) i5-10400F @2.90GHz
Learning rate	0.0001
Goal	-110
Min score	-200
Eps start	1
Eps end	0.001
Eps decay	0.9
Gamma	0.99
Batch size	64
Memory size	100000
Max episode	2000

Training record

Episode	Average score
100	-179.84
200	-179.21
300	-147.61
400	-147.29
500	-133.86
600	-127.31
700	-124.42
800	-119.81
900	-111.44

Results

Success at episode	911
Total average score	-106.32
Max reward	-84.0

Reward plot